

Day 6 29/11/2025

Loops

A loop is a programming structure "used to repeat a block of code multiple times"

- * If you want to execute something again and again instead of writing it many times you use a loop.

Types of loops

01 For loop

It is used when you know how many times you want to repeat something

Ex: ①

```
for i in range(5):  
    print('Hello')
```

②

```
name = 'chandana'
```

```
for char in name: # char is iterative variable  
    print(char)
```

output

c
h
a
n
d
a
n
a

02 while loop

It is used when you don't know the exact number of repetitions and continue until a condition becomes false.

```
Ex:- count = 1
      while count <= 5:
          print(count)
          count += 1
```

03 Nested loop

A nested loop means placing one loop within another loop.

* The inner loop runs completely every time the outer loop runs once.

```
Ex:- for i in range(3):
      for j in range(2):
          print(i, j)
```

Example problems

```
01 for i in range(1, 11, 1): # i = 1, 2, 3, ..., 10
    print(i, " = python")
```

output 1 = python

⋮

10 = python

02 for i in range(1, 11, 1):
print(i)

output 1

2

3

⋮

⋮

10

03 print Tables of given num from $n \times 1 = \text{ans}$
 $n \times 2 = \text{ans}$

⋮

$n \times 10 = \text{ans}$

=> n = int(input("enter a number"))
for i in range(1, 11, 1):
print(f"{n} * {i} = {n*i}")